Area between a curve and a line

Exam code: 9MA0

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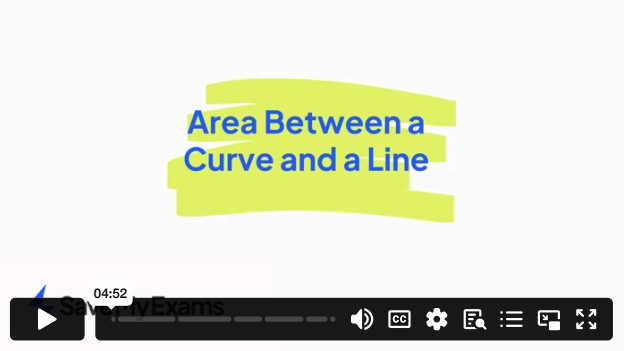
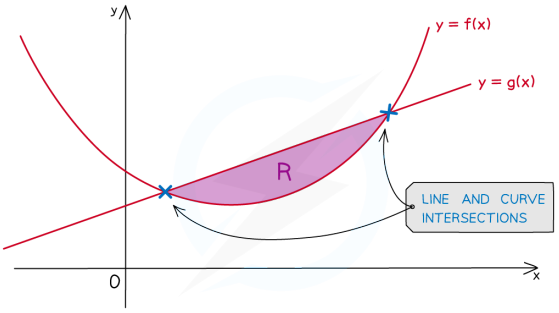


Did this video help you? Yes No

Area between a curve and a line

What is the area between a curve and a straight line?



R IS THE AREA BETWEEN A CURVE AND A STRAIGHT LINE

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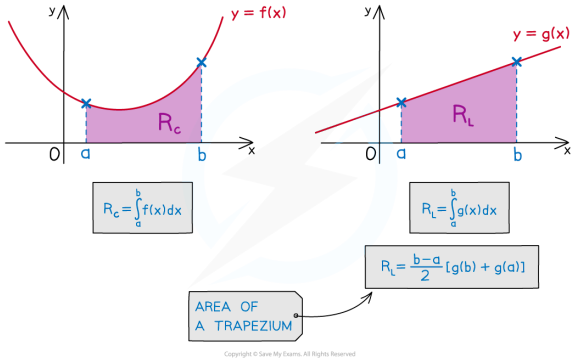
- This is the area enclosed by a curve and a line between their **intersections**
- The intersections may need to be found using simultaneous equations

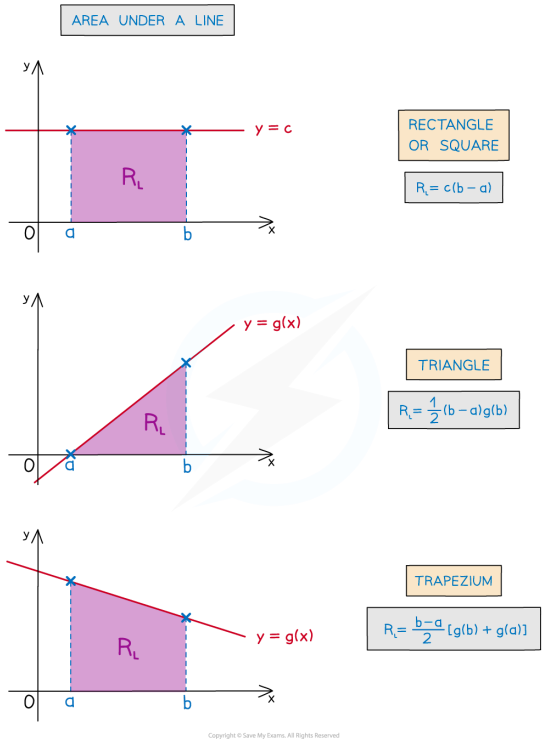
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

two areas?

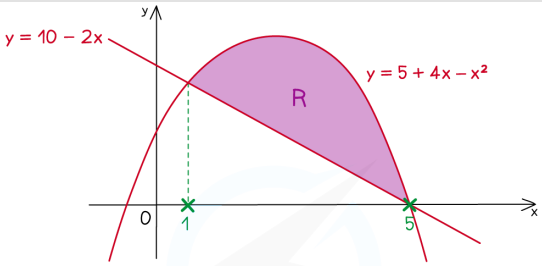
- The area enclosed will be the **difference** between ...
  - the **area under the curve** and
  - the **area under** the line
- These can be found separately



- There may be easier ways to find the area under a line



- Once the areas are found a question can be answered in full



STEP 1 FIND THE INTERSECTIONS OF CURVE AND LINE

$10 - 2x = 5 + 4x - x^2$

$x^2 - 6x + 5 = 0$

$(x - 5)(x - 1) = 0$

$x = 5, \quad x = 1$

SOLVE SIMULTANEOUSLY

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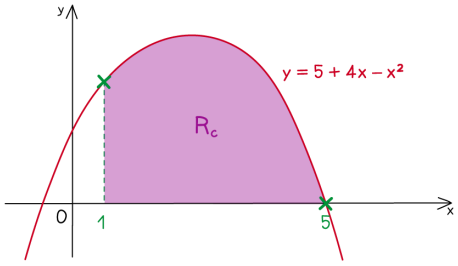
STEP 2 FIND AREA UNDER THE CURVE



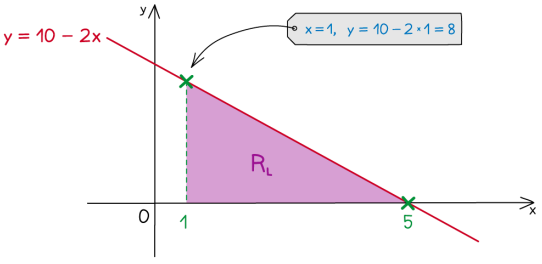
$R_c = \int_1^5 (5 + 4x - x^2)\,dx$

DEFINITE INTEGRATION

$R_c = \frac{80}{3}$

USE CALCULATOR

STEP 3 FIND AREA UNDER THE LINE



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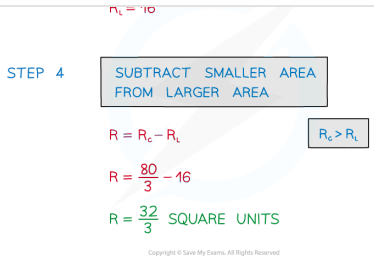
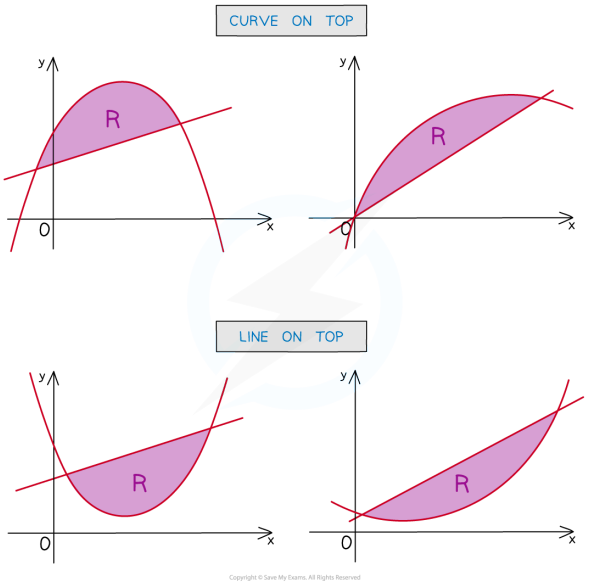


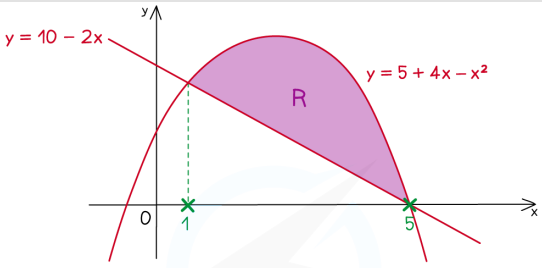
- STEP 1: Find the intersections of the line and the curve
- STEP 2: Find the area under a curve, $\mathbf{R_C}$, using definite integration
- STEP 3: Find the area under a line, $\mathbf{R_L}$, either using definite integration **or** the area formulae for basic shapes
- STEP 4: To find the area, **R**, between the curve and the line **subtract** the **smaller** area **from** the **larger** area
  - If curve on top this will be $\mathbf{R_C - R_L}$
  - If line on top this will be $\mathbf{R_L - R_C}$

How do I find the area between a curve and a line by subtracting their equations?

- An alternative method allows subtraction of the two functions first
- But be extremely careful as to whether the curve or the line is on top



- STEP 1: Find the intersections of the line and the curve
- STEP 2: Subtract the functions algebraically
  - Ensure this is done the correct way round!
- STEP 3: Find the area, **R**, between the curve and the line using **definite integration**



STEP 1 FIND THE INTERSECTIONS OF CURVE AND LINE

$$10 - 2x = 5 + 4x - x^2$$
$$x^2 - 6x + 5 = 0$$
$$(x-5)(x-1) = 0$$
$$x = 5, \quad x = 1$$

SOLVE SIMULTANEOUSLY

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STEP 2 SUBTRACT THE FUNCTIONS

$$5 + 4x - x^2 - (10 - 2x) = -x^2 + 6x - 5$$

CURVE ON TOP

SUBTRACT "ALL" OF LINE

STEP 3 DEFINITE INTEGRATION FINDS R

$$R = \int_1^5 (-x^2 + 6x - 5)\,dx$$

$$R = \frac{32}{3} \text{ SQUARE UNITS}$$

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Examiner Tips and Tricks

- Add information to any diagram provided
- Add axes intercepts, as well as intercepts between lines and curves
- Mark and shade the area you're trying to find
- If no diagram provided, **sketch** one!

Worked Example

Teacher only

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